Gravitational potential, particles and halos: GADGET-4 versus PKDGRAV3 from identical initial conditions

32^3 and 256^3 particles in $200\,h^{-1}\,\mathrm{Mpc}$, Sahyadri default settings on pawna

Sahyadri sandbox, branch `gadget-pkdgrav3-potential`

12 November 2026

Abstract

GADGET-4 and PKDGRAV3 were run from identical initial conditions (GADGET-4 NGenIC, Sahyadri default settings, $200\,h^{-1}\,\mathrm{Mpc}$) with 32^3 and 256^3 particles. Each run went through the full Sahyadri halo pipeline (Rockstar, consistent-trees, VAHC), and the two codes are compared particle by particle (potential, position, velocity), halo by halo, and through the stacked potential around halos in mass bins out to $50\,h^{-1}\,\mathrm{Mpc}$. At 256^3:

- **Particles.** At $z = 12$ the potentials agree to 6×10^{-3} rms (correlation $1 - 6 \times 10^{-6}$). The difference grows to 2% at $z = 3$ and 22% at $z = 0$, and 75–90% of it comes from the particles ending up in different places: at $z = 0$ the median separation is $0.12\,h^{-1}\,\mathrm{Mpc}$ and the median velocity difference $70\,\mathrm{km\,s^{-1}}$, dominated by halo particles. Once PKDGRAV3's softened self-potential is removed, the zero points agree to $\leq 1.2 \times 10^3\,(\mathrm{km\,s^{-1}})^2$.
- **Halos.** Mass functions agree within 2–3% above $10^{12}\,h^{-1}\,\mathrm{M_\odot}$. Matched halos have the same masses, spins and VAHC bias, but PKDGRAV3 halos are 20–25% more concentrated at 10^{12}–$10^{13}\,h^{-1}\,\mathrm{M_\odot}$, with 2–5% higher V_{max} and 4–9% deeper ϕ_{min}. This is consistent with PKDGRAV3's smaller effective softening under the production settings. For the same particles, the mean potentials of matched halos agree to 0.5%.
- **Stacked potential.** Around halos the stacked mean potential agrees to 1–2% ($\leq 0.02\,\sigma_\phi$) from 0.2 to $50\,h^{-1}\,\mathrm{Mpc}$ at $z = 3$, 1, 0.5 and 0. Inside $0.1\,h^{-1}\,\mathrm{Mpc}$ PKDGRAV3 is 3–6% deeper. The volume-weighted mean (potential on a 512^3 grid) is much shallower than the particle-weighted one and tends to zero far from the halos. Its code difference is within 2 paired errors ($\leq 0.016\,\sigma_\phi$) at every radius from 0.4 to $50\,h^{-1}\,\mathrm{Mpc}$, in both 36 log bins and 250 linear bins of $0.2\,h^{-1}\,\mathrm{Mpc}$.

The stacking tool keeps only per-mass-bin sums and dilutes the outer shells by particle ID. A stack of a 256^3 snapshot with $\sim 17\,000$ halos takes $\sim 10\,\mathrm{s}$; gridding the potential for the volume weighting takes 6 min.

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1 Set-up

Both codes evolve the *same* particle load. GADGET-4's NGenIC (2LPT, CLASS $P(k)$ at $z = 49$, the Sahyadri production recipe of `scripts/submit_CiPod.sh`) writes the initial conditions; the same file, converted to GADGET-2 format, is read by PKDGRAV3 (`achInFile`), so the two runs differ only in the N-body solver, the time stepping and the softening. Every setting is the Sahyadri default of `config/submit/sahyadri/sub_default.conf` unless stated (Table 1).

box, particles	$L = 200\,h^{-1}$ Mpc; $N = 32^3$ (test) and 256^3
cosmology	$\Omega_m = 0.3137721$, $\Omega_b = 0.0493017$ (GADGET-4 parameter only; both runs are dark matter only), $\Omega_\Lambda = 0.6862279$, $h = 0.6736$, $n_s = 0.9649$, $A_s = 2.098903 \times 10^{-9}$, seed 12345678
initial conditions	GADGET-4 NGenIC 2LPT at $z = 49$ (CLASS spectrum, no renormalisation)
outputs	Sahyadri's 100 output times (`config/sims/outputs100.txt`, $z = 12 \to 0.0093$) plus $a = 1$: 101
GADGET-4	commit 1e171a4, production Config + `EVALPOTENTIAL`; TreePM, PMGRID $= 2N^{1/3}$, `ErrTolForceAcc` 0.002, $\theta_{\max} = 0.75$; softening $\epsilon_G = L/(30N^{1/3})$ (Plummer-equivalent)
PKDGRAV3	`shadaba/pkdgrav3` `main` (binary built from 5766bb28; later commits are documentation); production `sahyadri2048.par` settings: $\epsilon_P = 1/(60N^{1/3})$ box units, 20 steps between outputs, output at the redshifts GADGET-4 actually wrote; `dCompPotStep=1` (potential in the `.pkc` files)
softening	256^3: $\epsilon_G = 26.0\,h^{-1}$kpc, $\epsilon_P = 13.0\,h^{-1}$kpc ($32^3$: 208 and $104\,h^{-1}$kpc); PKDGRAV3's force is Newtonian beyond $2\epsilon_P = \epsilon_G$, GADGET-4's (spline) beyond $2.8\,\epsilon_G$
halos	Rockstar (production template: 30 particles output, 20 minimum, unbound threshold 0.5; GADGET-4 HDF5 via `FILE_FORMAT = "AREPO"`, PKDGRAV3 via the native `.pkc` reader), consistent-trees, VAHC (grid 512^3)

Table 1: Simulation and pipeline settings.

Pipeline. One command submits everything as a PBS chain (`scripts/gadget_pkdgrav3_compare/submit_compare.sh compare256.conf`): GADGET-4 job (CLASS → NGenIC → IC conversion → run) → PKDGRAV3 `sim` → Rockstar → consistent-trees → VAHC for each code (the Sahyadri PKDGRAV3 pipeline, `scripts/pkdgrav3_pipeline`) → comparison job (`job_compare.pbs`). At 32^3 consistent-trees cannot run (at most 15 halos per snapshot; it needs enough halos to estimate its consistency tolerances), so the 32^3 halo results use the Rockstar catalogues.

2 Where things are

Data on pawna

Work directory `/storage/shadab/pkdgrav3_io/potcmp/` (below: `data/` = `/storage/shadab/pkdgrav3_io/potcmp/data/`).

what	path
GADGET-4 256^3 run	`data/sims/potcmp/gadget256/r1/`: `snapshot_000..100.hdf5` (with Potential), `param_run.txt`, `gadget_run.log`, `snapnames.txt`, `gadget_out_z.txt`
initial conditions 256^3	`data/sims/potcmp/gadget256/r1/ics/`: `snapshot_ics_000.hdf5` (GADGET-4), `ics_gadget2.dat` (PKDGRAV3 input), `param_ics.txt`; CLASS: `data/transfer/class_gadget256*`
PKDGRAV3 256^3 run	`data/sims/potcmp/pkd256/r1/`: `p.par`, `snap.<step>` (HDF5), `snap.<step>.s<pct>.0.pkc`, `snapnames.txt`, `run.log`
halos 256^3	`data/halos/potcmp/gadget256/r1/`, `data/halos/potcmp/pkd256/r1/`: `out_<snap>.list` (Rockstar), `out_<snap>.trees` (consistent-trees hlists), `out_<snap>.vahc`, `trees/`, `hlists/`; `../scales.txt`
comparison 256^3	`data/compare/pkd256_vs_gadget256_r1/`: `particles.csv`, `halos.csv` (potentials of matched halos), `summary.json`, `snap_*.npz` (200 000-particle subsamples), `profiles/{gadget,pkd,pkc}_out<i>.npz` (stacks, ID-hash dilution; the job's first version with per-chunk random dilution in `profiles_jobrun_target4096/`), `halos/` (`halo_mf.csv`, `halo_match.csv`, `halo_match_out<i>.npz`), `compare.log`
32^3 (same layout)	`data/sims/potcmp/{gadget32,pkd32}/r1/`, `data/halos/potcmp/{gadget32,pkd32}/r1/`, `data/compare/pkd32_vs_gadget32_r1/` (stacks in `profiles/`; the earlier $r/R_{\rm vir}$ profiles in `profiles_rvir_old/`)
first 32^3 run (no `EVALPOTENTIAL`)	`/storage/shadab/pkdgrav3_io/potcmp/data_run1_zeropot/` (Potential $= 0$; kept as evidence)
job logs	`data/logs/<stage>.potcmp_{gadget,pkd}256_r1.<job>.log`, chains `data/logs/chain.*.txt`
GADGET-4 builds	`/storage/shadab/pkdgrav3_io/potcmp/g4home/code/Gadget-4/{mesh512, mesh512-NGenIC,mesh64,mesh64-NGenIC}/Gadget4`
PKDGRAV3 binary	`/storage/shadab/pkdgrav3_io/build/pkdgrav3`

Scripts and report (repository `a-paranjape/sahyadri-sandbox`, branch `gadget-pkdgrav3-potential`)

`scripts/gadget_pkdgrav3_compare/`	
`compare32.conf`, `compare256.conf`, `*_gadget.conf`	run configurations (one per resolution; the `_gadget` file switches to the GADGET-4 side)
`submit_compare.sh`	submits the whole chain
`job_gadget.pbs`, `setup_gadget4_pawna.sh`, `Config.sh.template`, `make_gadget4_param.py`	GADGET-4 build, ICs and run, Sahyadri recipe
`gadget4_ic_to_gadget2.py`	the GADGET-4 ICs for PKDGRAV3 (IDs $\rightarrow$ iOrder, mass so that PKDGRAV3's total is Ω_m)
`job_compare.pbs`	comparison job: the three scripts below
`compare_potential.py`	particle by particle: potential, positions, velocities; FFT Poisson reference
`halo_potential_profile.py`	stacked potential around halos in mass bins (Section 3)
`compare_halos.py`	mass functions, matched halos, VAHC quantities
`reports/gadget_pkdgrav3_potential/`	this report: `report.tex`, `stack_potential.ipynb` (stacked potential figures), `make_figures.py` (other figures), `figs/`, `data/` (copies of the small comparison outputs: `particles.csv`, `profiles/`, `halos/*.csv`)

3 Methods

Units and zero point of the potential. All potentials are peculiar and physical, in $(\mathrm{km\,s^{-1}})^2$. GADGET-4: ϕ = `Potential`$/a$ (attribute `a_scaling` $= -1$; the self-interaction is excluded and the periodic potential has volume mean zero). PKDGRAV3: ϕ = `Potential`$\,U^2/a$ with $U = 100\,L/\sqrt{8\pi/3}\,\mathrm{km\,s^{-1}}$ ($G = 1$, $\rho_{\mathrm{crit}} = 3/8\pi$, $L = 1$); PKDGRAV3 includes each particle's softened self-interaction $-\frac{105}{64}\,m/\epsilon_P$, a constant for equal masses, which is removed. The two codes can still differ by a constant per snapshot (zero point of the periodic sum, e.g. the Madelung-like term of a particle's own images). It changes no force, so it is measured as the median of $\phi_{\mathrm{pkd}} - \phi_{\mathrm{Gadget}}$ over all particles and removed ("aligned"). The `.pkc` potential is PKDGRAV3's, quantised to a step of $1\,(\mathrm{km\,s^{-1}})^2$ (maximum error 0.5, checked at every output).

Velocities. Peculiar, $\mathrm{km\,s^{-1}}$: GADGET-4 `Velocities`$\sqrt{a}$, PKDGRAV3 `Velocities`$\,Ua$ (at $z = 12$ the two rms velocities agree to 0.4%).

Particles are matched by ID (GADGET-4 ID -1 = PKDGRAV3 iOrder). For every 5th output (every output at 32^3) `compare_potential.py` reports the correlation and rms difference of ϕ, $|\Delta\vec{x}|$ and $|\Delta\vec{v}|$ percentiles, and, as an independent reference, the potential from an FFT Poisson solution of each snapshot's own particles (CIC on $\min(4N^{1/3}, 512)^3$, deconvolved). The rms difference between the FFT potentials of the two codes' particle positions isolates the part of $\Delta\phi$ that comes from the particles being in different places rather than from the force solver.

Stacked potential profiles. `halo_potential_profile.py` computes, for host halos (`pid` $= -1$) in bins of M_{vir}, the mean $\langle\phi\rangle(r)$ and standard deviation $\sigma_\phi(r)$ of the potential of all (halo, particle) pairs in logarithmic shells of comoving distance r from 0.02 to $50\,h^{-1}\,\mathrm{Mpc}$ (36 bins), and the jackknife error of the mean over 3^3 sub-cubes of halo centres. Each code uses its own halos and its own particles. Only stacks are accumulated: per chunk of particles, a periodic KD-tree of the chunk is paired with trees of halo centres, and every pair adds $(1, \phi, \phi^2, r)$ to a (sub-cube, mass bin, shell) array, so memory does not depend on the number of halos or particles. To keep the cost per halo constant out to large radii the outer shells are diluted: shell i uses the nested subsample $u < 8^{-\ell_i}$ of the particles, with the smallest ℓ_i for which the expected number of particles inside the shell's outer radius is ≤ 16384. Here u is a hash of the particle ID (GADGET-4 ID -1 = PKDGRAV3 iOrder), uniform in $[0, 1)$ and independent of ϕ, so both codes use the same particles at every level and the subsampling noise cancels in their difference. (A first version drew u at random per chunk; its outer-shell noise, $\sim 2 \times 10^4\,(\mathrm{km\,s^{-1}})^2$, showed up as a spurious 3% code difference at $45\,h^{-1}\,\mathrm{Mpc}$ and as a difference between PKDGRAV3's HDF5 and `.pkc` stacks.) Each shell uses one level, so its mean and scatter are unbiased (checked against a brute-force calculation: identical without dilution, within $1.4\sigma_{\mathrm{jk}}$ with it). The pair counts also give the halo–matter cross-correlation $1 + \xi_{hm}(r)$. At 2048^3 the outermost level uses 8^{-5} of the particles, and the cost is $\sim n_{\mathrm{halo}}\times$ levels $\times 16384$ pairs plus one pass through the snapshot. The per-sub-cube sums are stored, so code differences can be given a *paired* jackknife error: leave the same sub-cube out of both codes and take the scatter of the difference. That removes the large-scale variance the two runs share, which dominates the error of each stack on its own. The figures are made by the notebook `stack_potential.ipynb`.

Volume-weighted stacks and linear bins. Counting every (halo, particle) pair makes the stack a *mass-weighted* average over the shell: dense parts (filaments, neighbouring halos) count more than voids. The *volume-weighted* mean and variance average over space instead (`--weight volume`). Each code's potential is put on an N_g^3 grid of cell centres (512^3, spacing $0.39\,h^{-1}\,\mathrm{Mpc}$, at 256^3) by a weighted local linear fit $\phi \approx a + \vec{g} \cdot \Delta\vec{x}$ to the 16 nearest particles. The grid points are then stacked exactly like particles: same trees, jackknife, and dilution by grid index, so both codes use the same points. A plain inverse-distance average is biased toward the denser side of the neighbourhood, i.e. too deep near structures: on a synthetic clustered test it biased the stacked mean

by up to 8%, against $\leq 1.8\%$ (median 0.4%) for the linear fit. The interpolation reads all particles into one KD-tree; beyond 512^3 particles it has to be done slab by slab (not yet implemented). Every stack is made with two binnings: 36 logarithmic bins from 0.02 to $50\,h^{-1}\,\mathrm{Mpc}$, and 250 linear bins of $0.2\,h^{-1}\,\mathrm{Mpc}$ from 0 to $50\,h^{-1}\,\mathrm{Mpc}$. Shells with fewer than 50 pairs or grid points are masked, and so are volume-weighted shells inside one grid cell, where the stack only sees the interpolation. Script: `run_stacks_local.sh` (run on the Mac from copies of the four snapshots).

Halo catalogues. `compare_halos.py` compares host-halo mass ($M_{\rm vir}$, M_{200b}) and $V_{\rm max}$ functions, and matches host halos one to one (mutual nearest centres within $0.5\,R_{\rm vir}$ of the GADGET-4 halo and $|\log_{10} M_{\rm pkd}/M_{\rm Gadget}| < 0.5$). For matched halos it reports, per mass bin, the median and 16–84% range of the ratios of $M_{\rm vir}$, M_{200b}, $V_{\rm max}$, concentration $R_{\rm vir}/r_s$, spin, shape and virial ratio, the centre offset and bulk-velocity difference, and the differences of the VAHC halo bias b_1 and tidal eigenvalues.

4 Results at 32^3

The 32^3 run is the first test of the chain (particle mass $2.1 \times 10^{13}\,h^{-1}\,\mathrm{M}_\odot$, at most 15 halos per snapshot). Table 2 and the grey lines of Fig. 3 summarise the particle comparison.

out	z	corr(ϕ)	$\frac{\mathrm{rms}\,\Delta\phi}{\mathrm{rms}\,\phi}$	FFT$_x$	offset $(\mathrm{km\,s^{-1}})^2$	med $\lvert\Delta x\rvert$ $h^{-1}\,\mathrm{Mpc}$	90% $\lvert\Delta x\rvert$ $h^{-1}\,\mathrm{Mpc}$	med $\lvert\Delta v\rvert$ $\mathrm{km\,s^{-1}}$	corr(v)
0	11.98	0.99965	0.026	0.082	+1392	0.016	0.029	6.3	0.9989
19	2.96	0.99972	0.024	0.103	+451	0.098	0.167	13.3	0.9980
46	1.00	0.99520	0.098	0.117	+308	0.205	0.364	17.9	0.9633
64	0.50	0.98510	0.173	0.147	−419	0.271	0.531	20.8	0.9123
100	0.00	0.96891	0.253	0.220	−2011	0.379	0.891	24.9	0.8151

Table 2: 32^3, particle by particle. FFT$_x$: rms difference of the FFT potentials computed from the two codes' particle positions, over rms ϕ. Offset: median $\phi_{\rm pkd} - \phi_{\rm Gadget}$ after removing PKDGRAV3's self-term (-1.9×10^7 at $z = 12$, $-1.4 \times 10^6\,(\mathrm{km\,s^{-1}})^2$ at $z = 0$).

- **Potential.** Until $z \approx 3$ the two potentials agree to 2–3% rms (correlation 0.9997). Afterwards the difference grows to 25% at $z = 0$; the FFT potentials of the two codes' own particle positions differ by 22%, so most of the late-time difference comes from the particles being in different places, not from the force solvers. The zero points agree to $\lesssim 2 \times 10^3\,(\mathrm{km\,s^{-1}})^2$ (0.1% of the self-term that PKDGRAV3 carries).

- **Positions and velocities.** The median particle separation grows from $16\,h^{-1}$kpc at $z = 12$ to $0.38\,h^{-1}\,\mathrm{Mpc}$ at $z = 0$ ($1.8\,\epsilon_G$, 6% of the mean interparticle spacing $6.25\,h^{-1}\,\mathrm{Mpc}$), the median velocity difference from 6 to $25\,\mathrm{km\,s^{-1}}$. The rms velocity difference ($314\,\mathrm{km\,s^{-1}}$ at $z = 0$) is dominated by particles in halos, where trajectories are chaotic: there the median is small but the tail long (Fig. 1). The rms velocities of the two runs differ by 0.4% at $z = 12$ and 1.7% at $z = 0$.

- **Halos.** Rockstar finds 13 host halos in GADGET-4 and 8 in PKDGRAV3 at $z = 0$; 4 are matched one to one (the others merge or split differently at the 30–40 particle limit). Matched masses agree to a few per cent.

- **Stacked potential** (Fig. 2). Around the 9 halos above $4 \times 10^{14}\,h^{-1}\,\mathrm{M}_\odot$ the mean potential of the two codes agrees within the jackknife errors from 1 to $50\,h^{-1}\,\mathrm{Mpc}$; inside $\sim 0.5\,h^{-1}\,\mathrm{Mpc}$ PKDGRAV3's is deeper, as expected from its smaller effective softening (PKDGRAV3's force becomes Newtonian at $1.0\,\epsilon_G$, GADGET-4's at $2.8\,\epsilon_G$). With 1–9 halos per bin these numbers are indicative only.

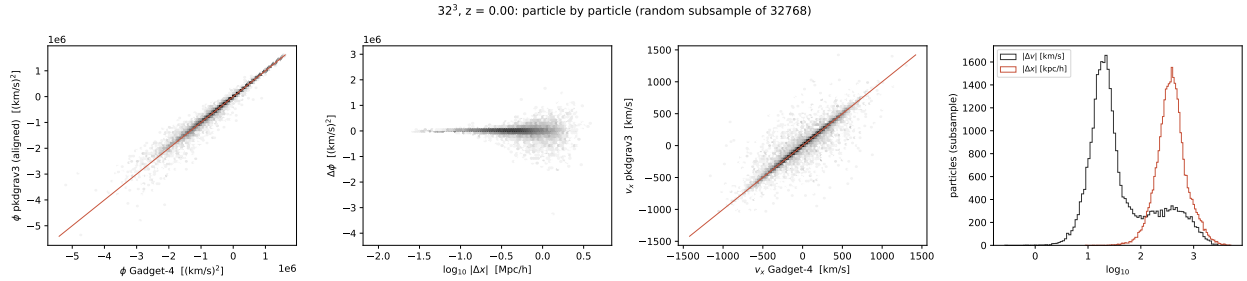


Figure 1: 32^3, $z = 0$, every particle: potentials (PKDGRAV3 aligned to GADGET-4's zero point), potential difference against particle separation, v_x, and the distributions of $|\Delta v|$ and $|\Delta x|$.

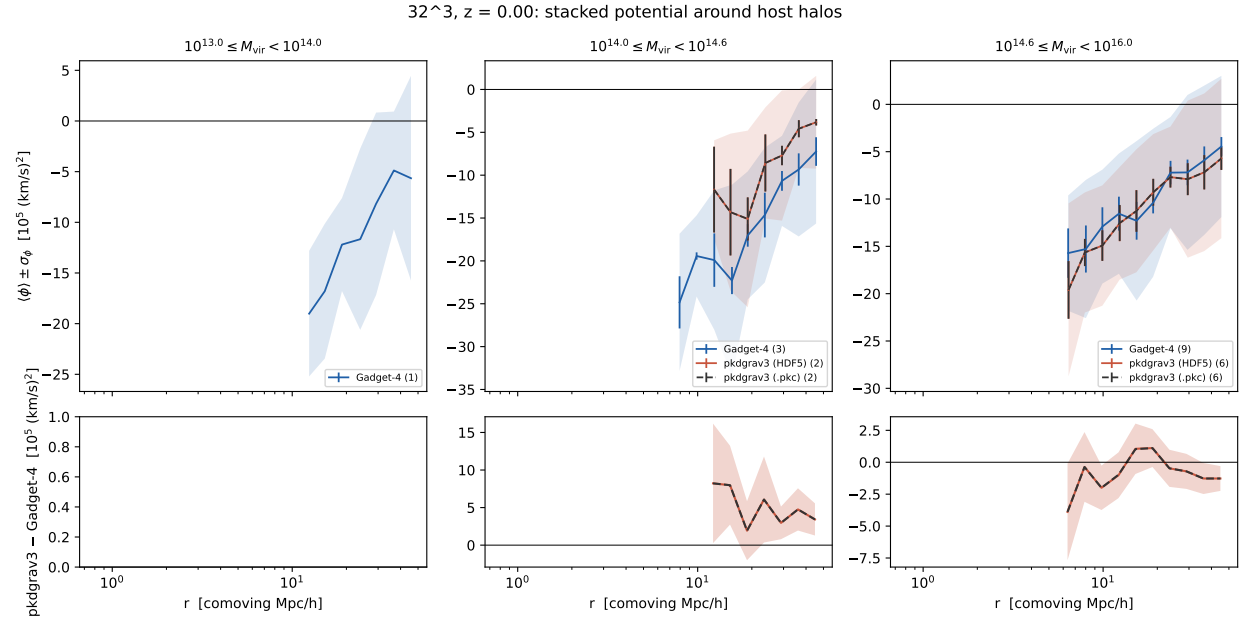


Figure 2: 32^3, $z = 0$: stacked potential around host halos in three $M_{\rm vir}$ bins (number of halos in the legend). Line with error bars: mean and jackknife error; band: $\pm\sigma_\phi$, the scatter of the potential over all pairs in the shell. Bottom: PKDGRAV3 − GADGET-4 (HDF5 and `.pkc` coincide).

The volume-weighted and linear-bin stacks were made at 32^3 too (64^3 grid, spacing $3.1\,h^{-1}$ Mpc). With 1–12 halos per bin and $0.2\,h^{-1}$ Mpc shells, almost every linear shell has fewer than 50 pairs and is masked. They are in `data/pkd32_vs_gadget32_r1/profiles` and the notebook, but carry no information beyond the 256^3 results.

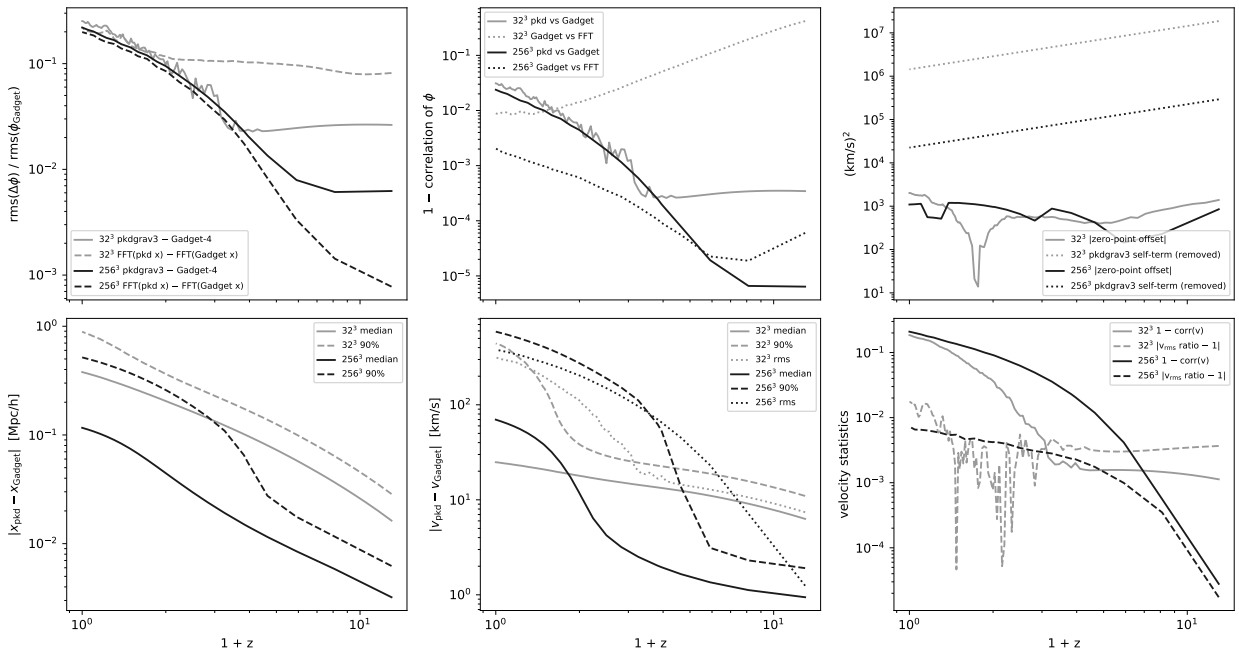


Figure 3: Particle-by-particle statistics against redshift (grey: 32^3, black: 256^3). Top: rms potential difference over rms potential (solid) and the same for the FFT potentials of the two codes' particle positions (dashed); 1−correlation of the potentials; zero-point offset and PKDGRAV3's removed self-term. Bottom: particle separation, velocity difference, velocity correlation and rms-velocity ratio.

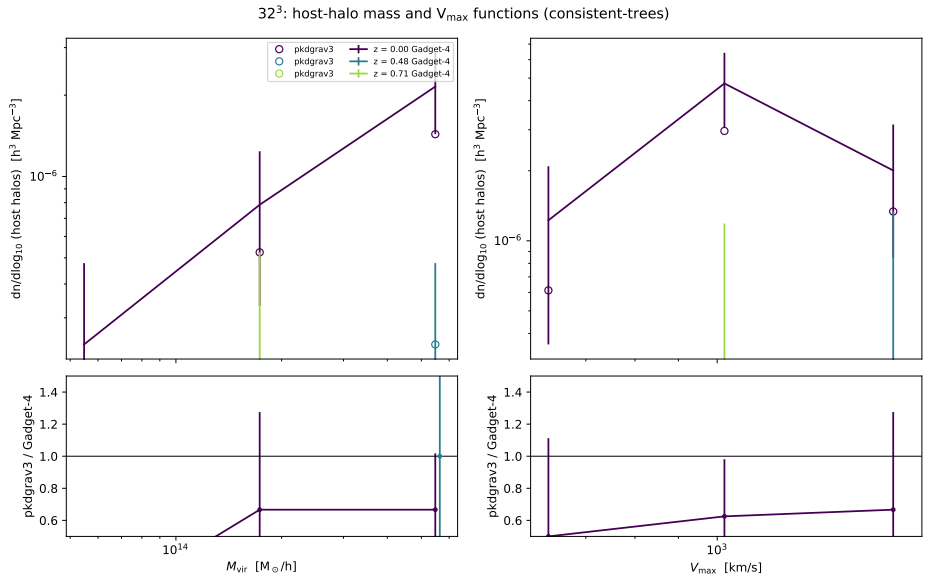


Figure 4: 32^3: host-halo $M_{\rm vir}$ and $V_{\rm max}$ functions (Rockstar) of both codes; with ≤ 15 halos per snapshot this is only a test of the tool.

5 Results at 256^3

The 256^3 run has particle mass $4.15 \times 10^{10}\, h^{-1}\,\mathrm{M}_\odot$, $\epsilon_G = 26\, h^{-1}$kpc and $\epsilon_P = 13\, h^{-1}$kpc. Both runs and the full halo pipeline completed (Table 3).

stage	GADGET-4	PKDGRAV3
simulation	2.9 h, 64 MPI ranks, 101 snapshots, 63 GB	2.6 h, 64 threads, 101 outputs, peak memory 14 GB
Rockstar	29 min, 40 658 halos at $z = 0$	29 min, 39 883 halos at $z = 0$
consistent-trees	10 min, 94 hlists ($z \leq 7.8$)	11 min, 94 hlists
VAHC (grid 512^3)	2.3 h with 8 processes (peak 168 GB)	3.5 h with 8 processes
comparison	29 min (23 outputs, peak 9 GB); stacks 3–13 s per code and output	

Table 3: 256^3 jobs on pawna. consistent-trees skips the 7 earliest catalogues (< 20000 bytes, production threshold), so at $z > 7.8$ the halo comparison uses the Rockstar catalogues.

5.1 Particles: potential, positions, velocities

out	z	corr(ϕ)	$\frac{\text{rms}\,\Delta\phi}{\text{rms}\,\phi}$	FFT$_x$	offset $(\text{km s}^{-1})^2$	med $\lvert\Delta x\rvert$ h^{-1} Mpc	90% $\lvert\Delta x\rvert$ h^{-1} Mpc	med $\lvert\Delta v\rvert$ km s^{-1}	corr(v)
0	11.98	0.999994	0.006	0.001	+849	0.0032	0.006	0.9	1.0000
19	2.96	0.999802	0.021	0.015	−640	0.014	0.055	2.0	0.979
46	1.00	0.995549	0.095	0.085	−966	0.045	0.259	11.9	0.908
64	0.50	0.990111	0.142	0.129	−1192	0.074	0.367	37.5	0.867
100	0.00	0.976212	0.220	0.199	−1093	0.116	0.517	69.8	0.791

Table 4: 256^3, particle by particle (columns as in Table 2). PKDGRAV3's removed self-term is $-2.9 \times 10^5\,(\text{km s}^{-1})^2$ at $z = 12$ and -2.25×10^4 at $z = 0$.

- **Before structure forms the two codes agree very closely.** At $z = 12$ the potentials correlate to $1 - 6 \times 10^{-6}$ with 0.6% rms difference. That is better than either code agrees with the FFT reference ($1 - 6 \times 10^{-5}$), so it is limited by the reference, not by the codes. Particles are $3\,h^{-1}$kpc apart ($0.12\,\epsilon_G$) and velocities agree to $0.9\,\text{km s}^{-1}$.
- **The difference then grows with nonlinear structure**, to 2% rms at $z = 3$, 10% at $z = 1$ and 22% at $z = 0$ (25% at 32^3). From $z \approx 3$ on, 75–90% of it is accounted for by the particles being in different places: the FFT potentials of the two sets of positions differ by 0.73 (at $z = 3$) to 0.90 (at $z = 0$) of the rms $\Delta\phi$ (Fig. 3). The force solvers themselves agree far better than the rms suggests (see also the matched halos below).
- **The zero points agree** to $\leq 1.2 \times 10^3\,(\text{km s}^{-1})^2$ after removing PKDGRAV3's self-term, about 0.1% of rms ϕ. The `.pkc` potential matches PKDGRAV3's HDF5 to $0.5\,(\text{km s}^{-1})^2$ (half the quantisation step) at every output.
- **Positions and velocities.** At $z = 0$ the median particle separation is $0.116\,h^{-1}$ Mpc ($4.5\,\epsilon_G$; 15% of the mean interparticle spacing) and the 90th percentile $0.52\,h^{-1}$ Mpc. Velocities split into two populations (Fig. 5):
  - particles outside halos, $|\Delta v| \approx 3\,\text{km s}^{-1}$;
  - particles inside halos, whose orbits are chaotic, $|\Delta v| \sim 100$–$300\,\text{km s}^{-1}$.

  The median is $70\,\text{km s}^{-1}$ and the rms $380\,\text{km s}^{-1}$. The rms velocities of the two runs differ by 0.7% at $z = 0$.

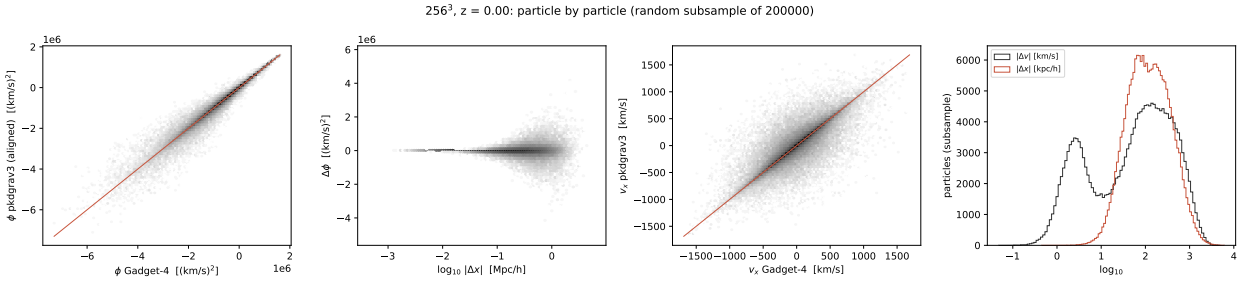


Figure 5: 256^3, $z = 0$, a random 200 000 particles: potentials, $\Delta\phi$ against separation, v_x, and the distributions of $|\Delta v|$ and $|\Delta x|$. The bimodal $|\Delta v|$ separates field particles from halo particles.

5.2 Halo catalogues and VAHC

At $z = 0$ there are 36 043 (GADGET-4) and 35 404 (PKDGRAV3) host halos; 28 484 are matched one to one.

- **Mass function** (Fig. 6). The $M_{\rm vir}$ functions agree within 2–3% (and within Poisson errors) from 10^{12} to $10^{14.5}\,h^{-1}\,{\rm M}_\odot$ at every redshift. Below $\sim 10^{12}\,h^{-1}\,{\rm M}_\odot$ (30 particles) PKDGRAV3 has 5% fewer halos at 5×10^{11} and 13% fewer at 2×10^{11}.

- **The $V_{\rm max}$ function differs more.** PKDGRAV3 has 14–17% fewer halos at 100–160 km s^{-1} and 13–19% more at 300–600 km s^{-1} ($z = 0$; about half as much at $z = 1$). This follows from the halo structure below.

- **Matched halos** (Fig. 7). The matched fraction is $\geq 96\%$ above $3 \times 10^{12}\,h^{-1}\,{\rm M}_\odot$, 79% at 10^{12}–3×10^{12} and 45% at 3×10^{11}–10^{12}.

  - *Masses* agree: median ratio 1.00 above $10^{12}\,h^{-1}\,{\rm M}_\odot$, with 16–84% range ±12% at 10^{12} narrowing to ±2% at 10^{14}.
  - PKDGRAV3 *halos are more concentrated*: $R_{\rm vir}/r_s$ is higher by 21–26% at 10^{12}–$10^{13}\,h^{-1}\,{\rm M}_\odot$, 9% at 3×10^{13} and 3% above 10^{14}. Their $V_{\rm max}$ is 2–5% higher.
  - *Shapes*: above $3 \times 10^{12}\,h^{-1}\,{\rm M}_\odot$ PKDGRAV3 halos are somewhat rounder, with c/a higher by 6–17%.
  - *Spin* agrees (median ratio 0.99–1.02).
  - *Centres and bulk velocities*: the median centre offset is $0.24\,R_{\rm vir}$ at 10^{12}, 0.11 at 10^{13} and 0.03 at $3 \times 10^{14}\,h^{-1}\,{\rm M}_\odot$. Bulk velocities differ by 15–80 km s^{-1} (median).

  The higher central density is consistent with PKDGRAV3's smaller effective softening ($2\epsilon_P = \epsilon_G$ against GADGET-4's spline support $2.8\epsilon_G$). It is largest at 10^{12}–$10^{13}\,h^{-1}\,{\rm M}_\odot$, where r_s is only a few ϵ_G.

- **VAHC.** The halo-by-halo bias b_1 is measured from the density field at $k < 0.1\,h\,{\rm Mpc}^{-1}$, which the two runs share. It is identical (correlation 1.0000; rms difference 0.010, against a halo-to-halo scatter of 3.4). The tidal eigenvalue λ_1 correlates at 0.98 with $R = 4R_{200b}$ smoothing (rms difference 1.0, scatter 5.4) and at 0.94 with $2R_{200b}$, where the halo's own surroundings, which differ in detail, matter more.

- **Potential of matched halos** (Fig. 8).

  - PKDGRAV3's $\phi_{\rm min}$ is 4–9% deeper (median per mass decade at $z = 0$), the mean potential of the 10% most bound particles 4–6% deeper, and $\langle\phi\rangle$ within $R_{\rm vir}$ 1–2% deeper. This is the same central-density difference.
  - For the same particles, PKDGRAV3's mean potential of each GADGET-4 halo's members agrees with GADGET-4's to 0.1–0.5%. That is the cleanest test of the two potentials, free of halo-finder and centring differences.

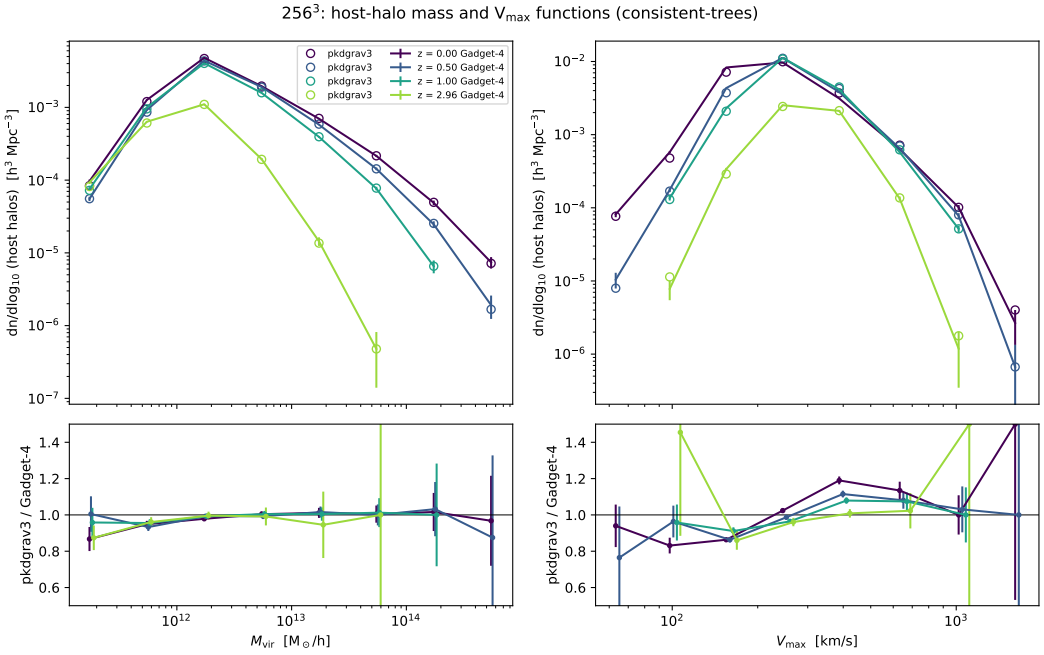


Figure 6: 256^3: host-halo $M_{\rm vir}$ and $V_{\rm max}$ functions from the consistent-trees catalogues of the two codes at $z \approx 3$, 1, 0.5 and 0 (lines with errors: GADGET-4; circles: PKDGRAV3). Bottom: ratio with Poisson errors.

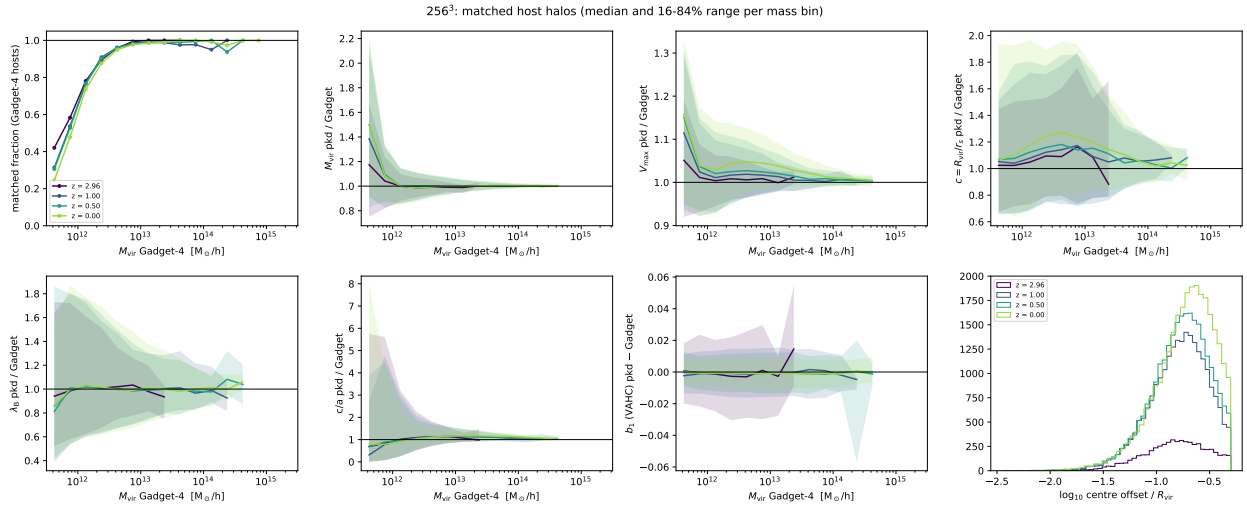


Figure 7: 256^3: matched host halos. Matched fraction, and per GADGET-4 mass bin the median (line) and 16–84% range (band) of PKDGRAV3/GADGET-4 for $M_{\rm vir}$, $V_{\rm max}$, concentration, spin and c/a; the difference of the VAHC bias b_1; and the distribution of centre offsets.

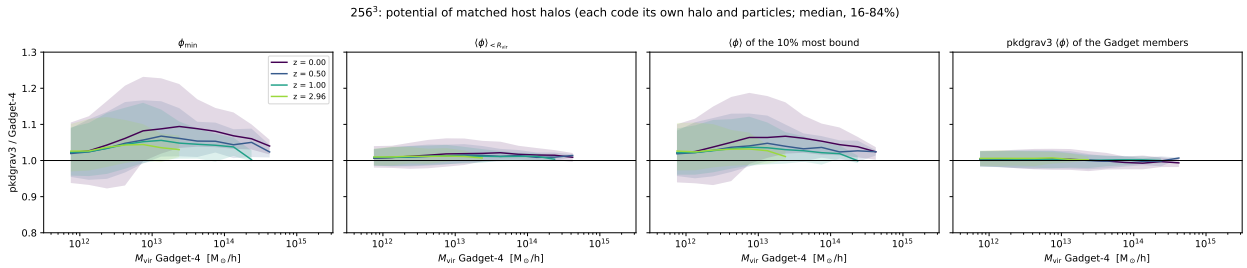


Figure 8: 256^3: potential of matched host halos, PKDGRAV3/GADGET-4. Each code uses its own halo and its own particles within $R_{\rm vir}$, except in the last panel, where PKDGRAV3's potential is evaluated for the GADGET-4 halo's member particles (same IDs).

5.3 Stacked potential around halos

Figures 9–12 come from the notebook `stack_potential.ipynb`. Each code uses its own host halos in four $M_{\rm vir}$ bins. At $z = 0$ these hold 13 375 / 3 181 / 525 / 78 halos in GADGET-4 (2×10^{12}–10^{13}, 10^{13}–5×10^{13}, 5×10^{13}–2×10^{14}, $> 2 \times 10^{14}\,h^{-1}\,{\rm M}_\odot$) and the same within 1% in PKDGRAV3. Differences carry the paired jackknife error: the two codes' differences over the same sub-cubes, which removes the shared large-scale (sample) variance. The dilution uses the same particle IDs in both codes, so the subsampling noise of the outer shells cancels as well.

- **Shape.** $\langle\phi\rangle(r)$ is deepest at the centre, from $-5 \times 10^5\,({\rm km\,s^{-1}})^2$ in the lowest bin to -5×10^6 in the highest. In the low-mass bins it has a second minimum at 1–3 h^{-1} Mpc: small halos often sit near larger ones. It is still $\approx -6 \times 10^5\,({\rm km\,s^{-1}})^2$ at $50\,h^{-1}$ Mpc, because halos live in large-scale potential wells. The scatter σ_ϕ in a shell, $(5\text{–}10) \times 10^5\,({\rm km\,s^{-1}})^2$, is set by the large-scale potential and is much larger than any code difference.
- **pkdgrav3 − Gadget-4.**
  - Inside $\sim 0.1\,h^{-1}$ Mpc PKDGRAV3 is deeper: by 3–5% at $0.05\,h^{-1}$ Mpc (3–6 paired errors), and by up to $3 \times 10^5\,({\rm km\,s^{-1}})^2$ (6%) at $0.02\,h^{-1}$ Mpc in the most massive bin. This is the softening and concentration difference.
  - From 0.2 to $50\,h^{-1}$ Mpc the mean potentials agree to 1–2% (mostly within 2 paired errors) at all four redshifts. In units of the shell scatter that is $|\Delta\langle\phi\rangle| \leq 0.02\,\sigma_\phi$.
- **`.pkc` = HDF5.** PKDGRAV3's compressed snapshots give the same stacks as its HDF5 snapshots at all radii.
- **Matter around the halos** (Fig. 12). $1 + \xi_{hm}$ agrees within 2% beyond $0.1\,h^{-1}$ Mpc. Inside, PKDGRAV3 has more mass: +50% at $0.02\,h^{-1}$ Mpc in the lowest bin, falling to $\lesssim 10\%$ in the highest.

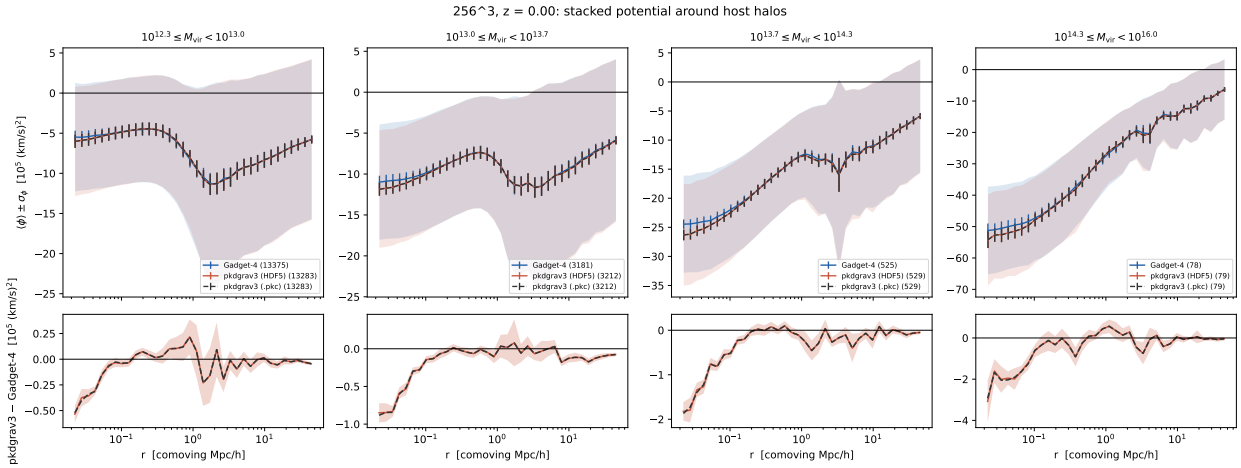


Figure 9: 256^3, $z = 0$: stacked potential around host halos in four $M_{\rm vir}$ bins out to $50\,h^{-1}\,$Mpc (number of halos in the legend). Lines with error bars: mean and jackknife error; bands: $\pm\sigma_\phi$. Bottom: PKDGRAV3 − GADGET-4 (solid: HDF5, dashed: `.pkc`) with the paired jackknife error.

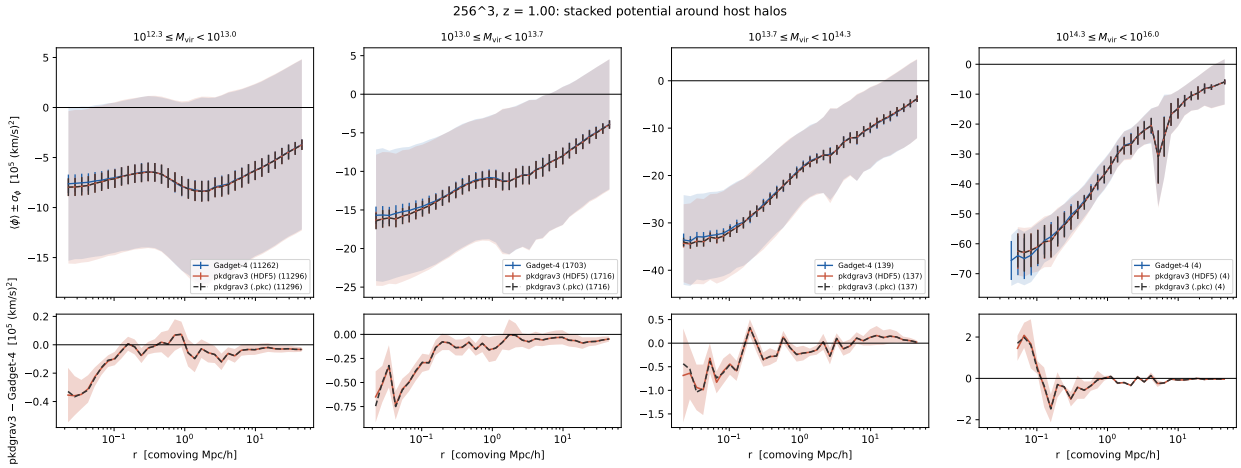


Figure 10: As Fig. 9 at $z = 1$.

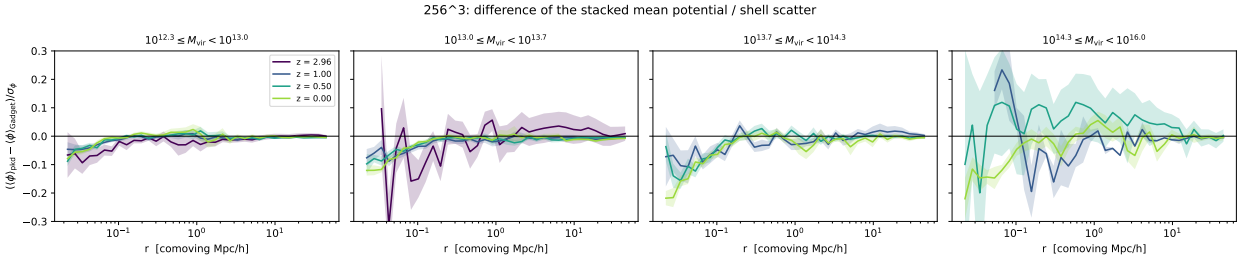


Figure 11: 256^3: $\Delta\langle\phi\rangle/\sigma_\phi$ at $z \approx 3$, 1, 0.5 and 0, with the paired jackknife error.

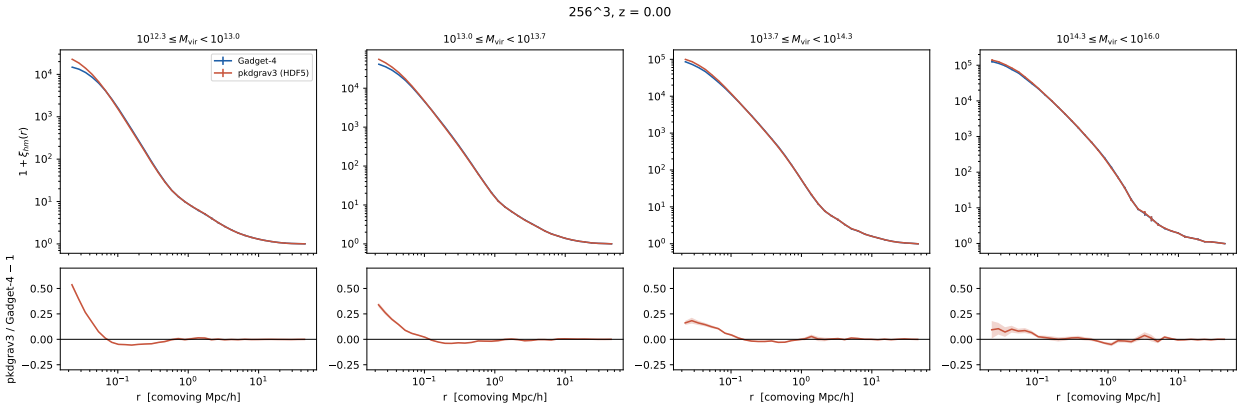


Figure 12: 256^3, $z = 0$: halo–matter cross-correlation from the same stacks; bottom: PKDGRAV3/GADGET-4 -1 with the paired jackknife error.

5.4 Volume-weighted stacks and linear bins

Figures 13–15 show the particle-weighted stacks (solid) and the volume-weighted stacks (dashed, 512^3 grid points) for both codes in the same mass bins. Table 5 lists them at $z = 0$ at selected radii of the $0.2\,h^{-1}$ Mpc linear bins. Volume-weighted shells inside one grid cell ($r < 0.39\,h^{-1}$ Mpc) are masked.

- **Volume versus particle weighting.** The volume-weighted mean is much shallower at every radius, and the gap grows outward. At $z = 0$ in the lowest bin it is $-3.1 \times 10^5\,(\mathrm{km\,s^{-1}})^2$ at $2\,h^{-1}$ Mpc (particle-weighted: -1.1×10^6) and -3×10^4 at $45\,h^{-1}$ Mpc (-5.7×10^5). Space far from a halo is mostly voids, where $\phi > 0$. The volume mean therefore tends to the box mean (ϕ averages to zero over the box), while particles keep sampling the wells of neighbouring structure. The volume-weighted scatter is also smaller, $(5.5\text{–}7) \times 10^5$ against $(7\text{–}13) \times 10^5\,(\mathrm{km\,s^{-1}})^2$, and it is almost flat in r. The bump of the particle-weighted stack at $1\text{–}3\,h^{-1}$ Mpc (neighbouring larger halos) is absent from the volume-weighted one, which rises smoothly.

- **pkdgrav3 − Gadget-4, volume-weighted.** For every mass bin with at least 10 halos, at all four redshifts and at all radii from 0.4 to $50\,h^{-1}$ Mpc, the difference is within 2 paired errors, and $|\Delta\langle\phi\rangle_V| \le 0.016\,\sigma_\phi$. The largest coherent offset is in the lowest bin at $z = 0$: PKDGRAV3 is shallower by 1.2–1.8 errors ($0.007\,\sigma_\phi$, about $4 \times 10^3\,(\mathrm{km\,s^{-1}})^2$) at all radii, which looks like a residual zero-point offset rather than a radial feature.

- **pkdgrav3 − Gadget-4, particle-weighted, linear bins.** The first shell (0–$0.2\,h^{-1}$ Mpc) holds the inner halo, and PKDGRAV3 is deeper there by 3–8 paired errors ($0.02\text{–}0.07\,\sigma_\phi$), as in the log bins. Beyond it most radii are within 2 errors. A few shells at 10–$45\,h^{-1}$ Mpc reach 3–5 errors, but they stay $\le 0.022\,\sigma_\phi$. The volume-weighted stacks do not show these excursions. They come from particles in neighbouring halos and filaments, whose positions differ between the codes (median $\Delta x = 0.12\,h^{-1}$ Mpc).

r [h^{-1} Mpc]	0.1	0.5	1.1	2.1	5.1	10.1	20.1	45.1
2×10^{12}–$10^{13}\,h^{-1}\,\mathrm{M}_\odot$ (13 375 halos)								
$\langle\phi\rangle_P$ [$10^5\,(\mathrm{km\,s^{-1}})^2$]	−4.9	−5.5	−9.4	−11.4	−9.3	−8.3	−7.0	−5.7
$\langle\phi\rangle_V$ [$10^5\,(\mathrm{km\,s^{-1}})^2$]	–	−3.5	−3.3	−3.1	−2.6	−2.0	−1.3	−0.34
Δ_P/err	−4.0	1.0	0.5	0.2	−0.2	−0.5	0.9	−2.9
Δ_V/err	–	1.8	1.4	1.6	1.7	1.8	1.8	1.2
10^{13}–$5 \times 10^{13}\,h^{-1}\,\mathrm{M}_\odot$ (3 181 halos)								
$\langle\phi\rangle_P$	−9.8	−7.4	−9.0	−11.4	−10.9	−9.1	−7.4	−5.8
$\langle\phi\rangle_V$	–	−6.2	−5.2	−4.6	−3.6	−2.8	−1.7	−0.47
Δ_P/err	−7.8	−0.8	0.4	−0.3	0.3	−2.4	−4.9	−3.2
Δ_V/err	–	−0.4	−0.8	−0.9	−0.8	−1.1	−0.3	−0.6
5×10^{13}–$2 \times 10^{14}\,h^{-1}\,\mathrm{M}_\odot$ (525 halos)								
$\langle\phi\rangle_P$	−21.7	−15.4	−12.4	−13.3	−11.9	−10.3	−8.4	−6.0
$\langle\phi\rangle_V$	–	−13.5	−10.1	−8.0	−5.9	−4.4	−2.7	−0.67
Δ_P/err	−7.0	0.5	−1.0	0.2	−1.0	−1.0	0.9	0.0
Δ_V/err	–	−0.8	−0.1	−0.4	0.0	0.1	0.4	0.8
$> 2 \times 10^{14}\,h^{-1}\,\mathrm{M}_\odot$ (78 halos)								
$\langle\phi\rangle_P$	−46.5	−34.6	−26.6	−21.1	−16.1	−13.4	−10.1	−6.1
$\langle\phi\rangle_V$	–	−30.6	−22.2	−16.3	−10.6	−7.6	−4.5	−1.2
Δ_P/err	−2.8	−0.5	1.8	0.9	−0.4	0.3	−0.2	−1.7
Δ_V/err	–	−0.6	0.1	0.2	−0.1	0.0	0.9	0.4

Table 5: 256^3, $z = 0$, linear $0.2\,h^{-1}$ Mpc bins (shell centres): GADGET-4 particle-weighted (P) and volume-weighted (V) mean potential, and PKDGRAV3 − GADGET-4 over its paired jackknife error. The full table for all redshifts is printed in `stack_potential.ipynb`.

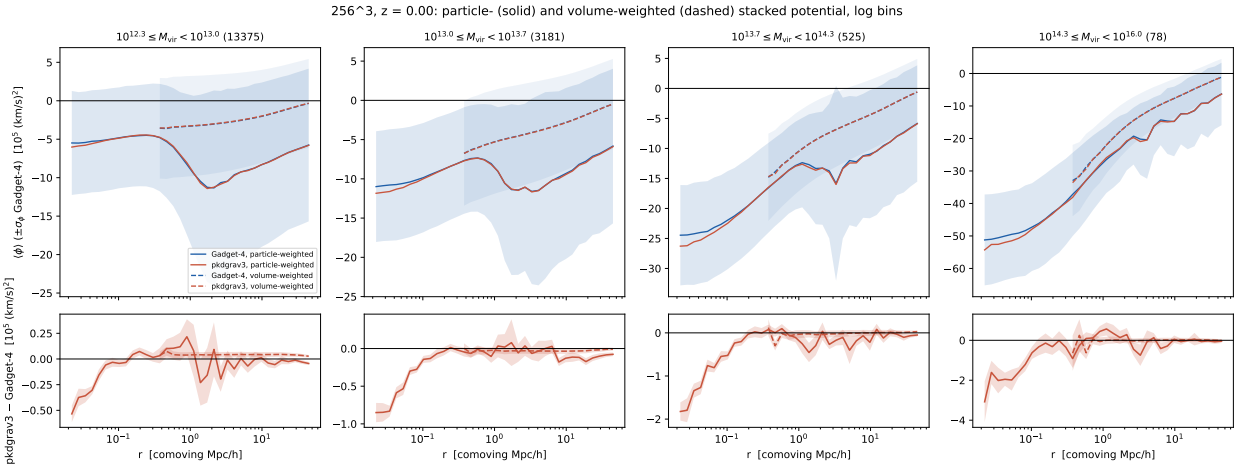


Figure 13: 256^3, $z = 0$, log bins: particle-weighted (solid) and volume-weighted (dashed) stacks for GADGET-4 and PKDGRAV3. Bands: $\pm\sigma_\phi$ of GADGET-4 for each weighting (the narrower band, starting at $0.4\,h^{-1}$ Mpc, is volume-weighted). Bottom: PKDGRAV3 − GADGET-4 with the paired jackknife error.

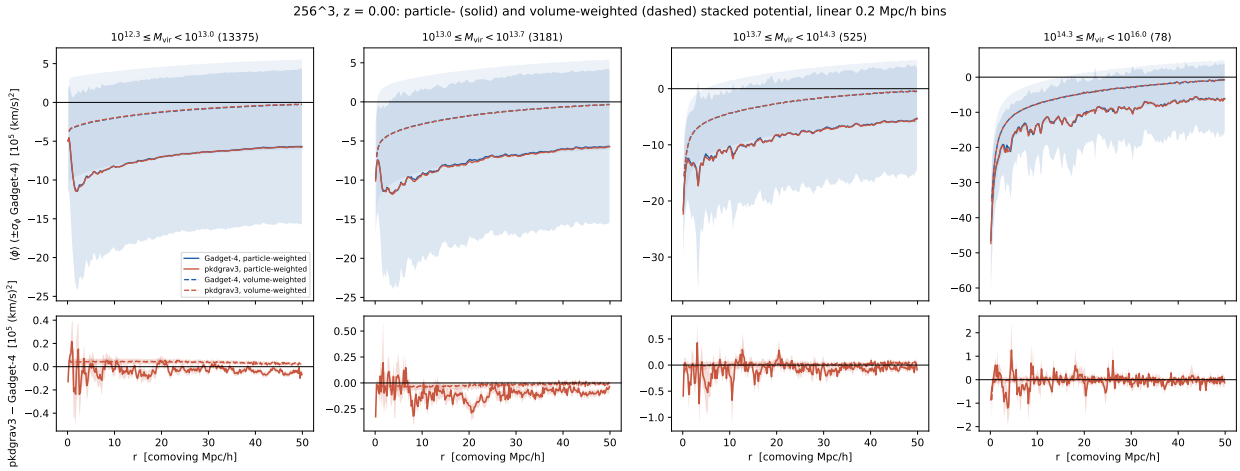


Figure 14: As Fig. 13 in 250 linear bins of $0.2\,h^{-1}$ Mpc from 0 to $50\,h^{-1}$ Mpc.

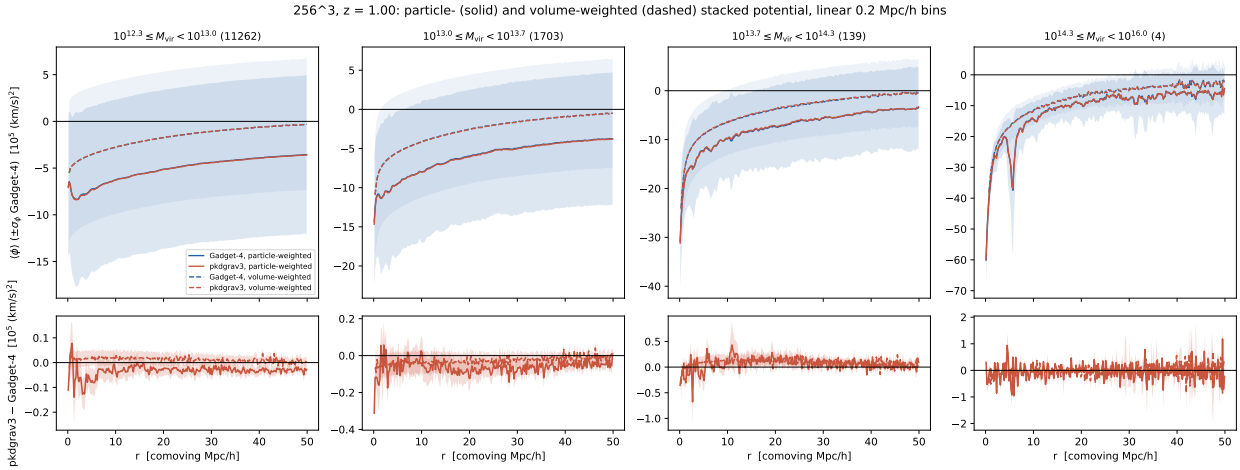


Figure 15: As Fig. 14 at $z = 1$.

6 Problems found on the way

1. **Gadget-4 writes Potential = 0 without `EVALPOTENTIAL`.** The first 32^3 run, with the exact production Config (`OUTPUT_POTENTIAL` only), has zero potential everywhere (`data_run1_zero pot/`). The production Sahyadri Config lacks `EVALPOTENTIAL`, so the production GADGET-4 snapshots most likely have no potential (not checked directly). Adding the flag left the ICs and output times byte-identical.

2. **consistent-trees needs many halos per snapshot**: it fails at 32^3 ("too few halos at scale factor 0.768278 to calculate consistency metric") even after relaxing `gen_merger_cfg.pl`'s 20000-byte catalogue threshold. At 256^3 the production threshold is used.

3. **VAHC is the slowest stage at 256^3**: 20–40 min per snapshot and process at grid 512^3 (FFT smoothing at many scales and per-halo loops), ∼23 GB per process. With 4 processes, 94 snapshots did not fit in 10 h; the jobs were rerun with 8 processes and 30 h (restartable: finished snapshots are skipped). At 2048^3 with 100 outputs this stage needs planning.

4. **Column names**: consistent-trees hlists call the columns `Mvir`, `Rvir`; Rockstar lists `ID`, `X`, ... The comparison scripts read both (fixed during the 256^3 run; the first two comparison jobs stopped on this).

5. **pkdgrav3 has no physical units when it reads an IC file** (`units.cxx`): box size in the output headers and `.pkc` potential units are wrong unless `dKpcUnit` and `dMsolUnit` are given; the pipeline's parameter generator now writes them. A Gadget-2 input file's IDs become iOrder directly, so they must be $0 \ldots N-1$; PKDGRAV3's HDF5 `Header/Time` is code time, not a. Recorded in the PKDGRAV3 fork, `docs/compressed_io_plan.md` section 7.

7 Reproducing

```
# on pawna, in a clone of sahyadri-sandbox (branch gadget-pkdgrav3-potential)
cd scripts/gadget_pkdgrav3_compare
bash -lc "bash setup_gadget4_pawna.sh <gadget4 source> <G4HOME> 256"   # Gadget-4 builds (onc
bash submit_compare.sh compare256.conf                                 # 9 PBS jobs
# figures (Mac or anywhere with matplotlib), from reports/gadget_pkdgrav3_potential:
rsync pawna:<data>/compare/pkd256_vs_gadget256_r1/{particles.csv,profiles,halos} data/pkd256_
jupyter nbconvert --to notebook --execute stack_potential.ipynb
python make_figures.py 32=data/pkd32_vs_gadget32_r1 256=<copy with snap_*.npz>
latexmk -pdf report.tex
```